

LESSON 110 (0.7 GROUND and 1.3 DUAL)**Lesson Objective**

The purpose of this lesson is to introduce normal/crosswind takeoffs and landings to allow the student to put together all of the fundamental maneuvers he/she has learned up to this point. This lesson will also introduce the student to climbs at different airspeeds (V_X and V_Y) as well as turning while climbing and descending. The student will also be introduced to the aiming point descent technique and Go-around/rejected landing. The Instructor will also conduct a power off stall demonstration and glide path demonstration. The student will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed.
 - Chapter 8 of the Airplane Flying Handbook - Airport Traffic Patterns.
 - Ask the students if they have any questions on the procedures from the maneuver's manual.
 - Normal/crosswind takeoff.
 - Power-off stalls.
 - Traffic pattern. (towered airport)
 - Go-arounds.
 - Normal/crosswind landing.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
 - Non-Towered Airport Flight Operations. ([AC 90-66](#))
- Aeronautical Information Manual.
 - Chapter 04 - Air Traffic Control.
 - Chapter 07 - Safety of Flight.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Warrior Maneuvers Manual.

Lesson Content (Ground)**Normal and Crosswind Takeoff**

- Taxi into takeoff position when cleared for takeoff by ATC and verify magnetic heading matches assigned takeoff runway heading.

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- Smoothly and continuously apply takeoff power.
- Check static RPM (approximately 2330 RPM) and verify airspeed rising.
- Use rudder as necessary to maintain directional control and ailerons (into the wind) to aid with crosswind correction.
- When the flight controls become effective, gradually apply back stabilator pressure to attain a lift-off pitch attitude. Ensure that ailerons are neutral prior to rotating.
- Note the position of the airplane's nose in relation to the horizon and adjust as necessary to attain a VY climb pitch attitude.
- At 1,000 feet AGL, lower the nose, approximately an inch, to attain a cruise climb pitch attitude and complete the climb check.

Climbing, Descending and Turning Techniques

- Review the lesson plan for lessons 108 and 109 as needed.

Traffic Pattern

- Draw and label the basic structure for a standard traffic pattern.
 - Power settings.
 - Airspeeds.
 - Non-specific vs specific aiming points.
- Explain the differences between standard and non-standard traffic patterns.
 - Direction of turns.
 - Altitudes.
 - Shape. (noise abatement procedures, high volume traffic)
- Entry.
 - Towered vs non-towered airports.

Stabilized Approach

- Define what a stabilized approach is.
 - Consistent and safe glide path.
 - Appropriate airspeeds.
 - Follows an extended centerline down to the runway.
- An unstable approach must result in a go-around.

Stall/Spin Awareness

- Define what an aircraft upset is:
 - "Unusual attitude" or loss of "positive control".
 - Use "Chapter 5 - Maintaining Aircraft Control..." in the AFH.
- Define what a stall is:
 - Differentiate aerodynamic stalls from engine stalls. (Overuse of abstraction).
 - Rapid decrease in lift caused by the separation of airflow from the wings surface when exceeding the critical AoA.
 - Wing does not stop producing lift – it cannot generate adequate lift to sustain level flight.

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- Can occur at any pitch, any attitude or airspeed.
- Describe in general terms how a stall might occur, and how to correct for it.
- Define what a spin is.
- Describe in general terms what a spin is, and the common phase of flight a spin might occur.

LAHSO, CFIT, Wire Strike Avoidance, Aviation Security and SRM

- Review Land and Hold Short Operations (LAHSO).
 - Who can accept it?
 - Do you have to accept it?
 - When can you deviate after accepting?
 - Show an example on an airport diagram (KSRQ, KPBI, KTPA).
- Review Controlled Flight Into Terrain (CFIT).
 - Unintentional collision with terrain (ground, mountain, water, or obstacle).
 - Commonly occurs on approach or during the landing phase of a flight.
 - How does it happen?
 - Loss of situational awareness.
 - Distractions.
 - Procedural mistakes (descent below MDA while IFR).
 - Aircraft performance (high density altitude, tailwind approaches, etc.).
 - How to avoid CFIT?
 - Read AIM 7-6-3 and AIM 7-6-4.
 - Flight Risk Assessment Tool (FRAT).
 - PAVE and IMSAFE acronyms.
 - Using all available resources.
 - Staying vigilant to changing conditions.
- Review wire strike avoidance.
 - AIM 7-6-3 dictates that avoidance of guy wires is very difficult due to the visibility of said wires
 - When flying at or below the altitude of antenna towers, the horizontal separation should be no less than 2000 feet.
- Review aviation security.
 - Ensure that the aircraft is properly secured after a flight:
 - Tie downs are on and tight.
 - Door and storm window closed.
 - Baggage compartment closed.
 - If you see something, say something!
 - AOPA airport watch: <https://www.aopa.org/advocacy/airports-and-airspace/security-and-borders/airport-watch-security>
 - Hotline:
 - 866-427-3287.
 - This provides a fast and easy way to report possible threats.

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- Review Safety Resource Management (SRM).
 - This starts before the preflight stage.
 - IMSAFE, being well acquainted with aircraft systems, etc.
 - Know what you are getting into.
 - Have set personal minimums prior to activity start.
 - Ensure the rewards outweigh the risks.
 - Understand the resources available to ensure a safe flight.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)**Preflight Inspection**

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal/Crosswind Takeoff

- Have the student vocalize the runway line-up checklist and verify all line items.
- Ensure the student establishes full power with one hand on the throttle and one hand on the yoke.
- Verify the student is inputting proper crosswind correction.
 - Ensure the student points out the windsock and corrects accordingly.
- Once off the ground, TRIM for desired airspeed!

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Climbing at Different Speeds (V_X , V_Y and Cruise Climb)

- Climbs are performed at a full power setting.
- Different airspeed climbs are established through changing the pitch attitude.
 - An increase in pitch will result in a decrease in airspeed.
 - A decrease in pitch will result in an increase in airspeed.
- Ensure the 1000' check is performed.
- Provide guidance for the student through climbing at V_X , V_Y , and cruise climb.
 - Have the student note the pitch attitude.
 - TRIM.

Climbing Turns

- Situational awareness and proper clearing procedures are critical.
- As angle of bank increases the vertical component of lift is reduced, thus causing the nose to drop.
- To compensate for this, as a turn is made, an increase in back pressure will be required to maintain the climb.
- TRIM.

Effect of Power and Flaps

- As flaps are lowered, drag increases, requiring an increase in power to maintain straight and level.
- As flaps are retracted, drag decreases, requiring a decrease in power to maintain straight and level.

Descending Turns

- Situational awareness and proper clearing procedures are critical.
- When descending over 100 feet, power should be brought to idle.
- With power idle, pitch controls airspeed much like with full power.
 - Increase in pitch results in a decrease in airspeed.
 - Decrease in pitch results in an increase in airspeed.

Non-Specific Aiming Point Descent Technique

- This is simulated by the downwind and base portions of the approach descent.
 - Pitch controls airspeed.
 - Power controls rate of descent.

Specific Aiming Point Descent Technique

- This is simulated by the final portion of the approach descent.
 - Pitch controls aiming point.
 - Power controls airspeed.
- If the airplane gets too low on the glidepath, add power, and increase pitch.
- If the airplane gets too high on the glidepath, decrease power, and decrease pitch.

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- The instructor should talk through a full power off stall maneuver, explaining how they are reacting to the aircraft.

Normal Traffic Pattern

- Conduct a rectangular course.
 - Downwind: 2500 ft.
 - Go-around: 1500 ft.
- Provide guidance for the student through the legs of the pattern.
 - Power settings.
 - Airspeeds.
 - Aiming points.
 - Crosswind correction.

Go-Around/Rejected Landing

- After conducting the rectangular course, have the student conduct a go-around.
- Provide guidance for the student through the go-around procedures.
 - Emphasize the importance of not going directly to flaps 0.
 - Emphasize the importance of adding power first.
- Ensure that the student uses TRIM while climbing out.

Normal/Crosswind Landing

- Instructor conducts the landing.
- Provide guidance for the student through the procedure, control inputs, and visual cues.
- Have the student follow along on the controls to gain a deeper understanding of what the instructor is doing.
- Have the student pay attention to the sight picture, and placement of the aiming point on final.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 111.
- Assign homework:
 - Read chapter 5 of the Airplane Flying Handbook - Maintaining Aircraft Control: Upset Prevention and Recovery Training.

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- Review the Warrior Maneuvers Manual:
 - Power off stalls.
 - Maneuvering during slow flight.

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Demonstrate climbing and descending turns with instructor guidance.
4. Maintain V_X and V_Y climbs with instructor guidance.
5. Conduct traffic patterns, takeoffs and landings with instructor guidance.
6. Describe the elements of a stabilized approach and determine when a Go-around/rejected landing is required.